

# Chinchilla Scaling Laws for Tiny Transformers

Hyeonseok Jung

## Abstract

I train transformers ranging from 5K to 3M parameters on 6-digit multiplication and derive compute-optimal scaling laws for them, following the three approaches of Hoffmann et al. (2022). All three approaches recover the Chinchilla conclusion that parameters and data should grow in roughly equal proportion with compute, with fitted exponents between 0.49 and 0.57 for the dense family. A mixture-of-experts family scales its active parameters at the same rate, and the benefit of sparsity appears in the constant of the frontier. The main obstacle was grokking. Above a compute threshold, runs fall discontinuously below every smooth fit. This behaviour dictated the choice of task, the fitting windows, and the interpretation of the fitted loss floor.

## 1 Problem statement

This report is about teaching small transformers 6-digit multiplication and deriving Chinchilla scaling laws from the resulting training runs. Following Hoffmann et al. (2022), I assume power-law frontiers

$$N_{\text{opt}}(C) \propto C^a, \quad D_{\text{opt}}(C) \propto C^b, \quad (1)$$

where  $C$  is the training compute budget,  $N$  the number of parameters, and  $D$  the number of training tokens. I estimate  $(a, b)$  with the paper’s three approaches, for a dense model family and for a mixture-of-experts (MoE) family.

### 1.1 Why 6-digit multiplication

The exercise this report started from proposes up-to-3-digit addition. I implemented that first and found that it cannot support a scaling analysis. At the larger step counts and FLOP budgets, every model in the sweep, including the smallest, saturated the task and reached 100% accuracy on the answer digits. Past that point the loss keeps falling, but it no longer measures ability. Once a model predicts every digit correctly, the remaining cross-entropy only reflects how much probability the softmax assigns to the already-correct answer, and it is driven by the sharpening of the logits over training rather than by anything learned about addition. Loss differences between saturated models therefore do not rank them. The behaviour below saturation was also unsuitable for fitting. Models sat near the digit-statistics loss and then *grokked*, jumping to the solution, and smaller models often jumped before larger ones. Chinchilla’s framework assumes a loss that falls smoothly with compute and that remains a meaningful measure of ability over the whole measured range, so that a frontier can be fit through it. I therefore made the task harder until the still-learning regime filled the compute range I could afford. With 6-digit multiplication the jump does not appear until roughly  $7 \times 10^{14}$  training FLOPs, which leaves about three decades of clean power-law behaviour. All fits below use that window. The transition itself is measured in Appendix A.

### 1.2 Task

Each example is a single 26-token sequence `lhs*rhs=answer` over a 12-token vocabulary (digits 0–9, \*, =). Every number is zero-padded to a fixed width, so both operands are always exactly 6 digits, the answer is always exactly 12 digits, and all sequences share one layout. Loss and accuracy are computed only on the 12 answer digits. Each batch is sampled uniformly at random from the  $10^{12}$  possible operand pairs, and a run sees at most  $10^9$  examples, so duplicates are rare and the sampler never runs out of fresh data. The training loss can therefore be used as the test loss, which is the same infinite-data assumption as in the paper (their footnote 2).

### 1.3 Models

The models are pre-RMSNorm decoder-only transformers with grouped-query attention, QK-norm, RoPE, and SwiGLU MLPs, computed in bf16. The dense family spans 55 sizes from 4.8K to 3.17M parameters. The MoE family replaces every MLP with a top-1-of-4-experts layer and covers 13 sizes from 20K to 2.98M *active* parameters. Precise sizings and hyperparameters are listed in Appendix F. Appendix B reports ablations of RoPE and QK-norm at the largest size.

### 1.4 Training

All models are trained with AdamW at batch size 256. The initial learning rate is set per model and follows a cosine schedule that decays by a factor of 10 over the full run. The loss is logged at every step. Two sweep protocols feed the analysis (Table 1).

| sweep         | protocol                                                                                   | dense      | MoE        |
|---------------|--------------------------------------------------------------------------------------------|------------|------------|
| fixed-horizon | each size $\times$ horizons of 4k–256k steps                                               | 257        | 52         |
| fixed-budget  | 8 budgets, $C = 1.8 \times 10^{13} \dots 3 \times 10^{15}$ , curated size range per budget | 206        | 67         |
| total         |                                                                                            | <b>463</b> | <b>119</b> |

**Table 1:** Training runs feeding the analysis.

### 1.5 Compute accounting

Instead of the  $6ND$  approximation, forward FLOPs per token are computed exactly for each architecture, including every matmul, normalisation, activation, and residual addition, and multiplied by 3 to account for the backward pass (Appendix C). The difference is not negligible at these scales; the exact count exceeds the  $6N$ -per-token rule by 17% for the smallest model and by 3% for the largest. All budgets  $C$  in this report are training FLOPs under this accounting.

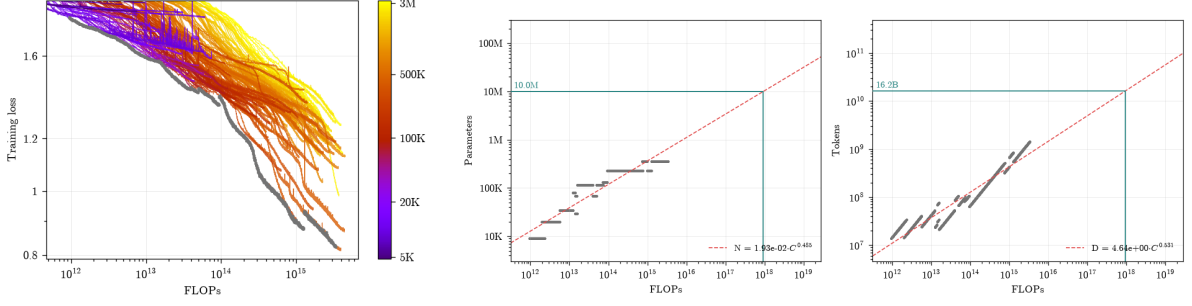
## 2 Approach 1: fix model sizes and vary number of training tokens (dense)

In the first approach I vary the number of training steps for the fixed family of model sizes. Each size is trained for four separate horizons of 4k, 8k, 16k, and 32k steps, a range of  $8\times$  in seen tokens, and subsets of the family are trained further at 64k, 128k, and 256k steps to extend the high-compute end. The cosine schedule decays the learning rate by a factor of 10 over each horizon, so a longer horizon is a separate run rather than a continuation. For each run I smooth the loss curve with a Gaussian filter ( $\sigma = 20$  steps) and interpolate it, which gives a continuous mapping from FLOP count to training loss. At 1,500 log-spaced FLOP values I then determine which run achieves the lowest interpolated loss and record its model size and token count. Finally, I fit power laws through these points to estimate the optimal model size and token count for any budget (Figure 1).

The training curves in Chinchilla are smooth enough that the raw envelope can be used directly. Mine are noisier, so I truncate the envelope where fewer than 10 distinct model sizes still reach the budget, which cuts the fit at  $C = 3.4 \times 10^{15}$ . Past that point a budget is won by whichever run happens to be the only one there, and those lone winners are worse per FLOP than the frontier at smaller budgets. One validity check from the paper also fails and should be noted as a caveat. Chinchilla report that every envelope point comes from the final 15% of a run (their footnote 4). In my data only 17.6% do, 62% come from the final half, and the median winning point sits at 60% of its run. This suggests that at this scale the envelope is won mid-run much more often than in their setting.

The fitted frontier is

$$N_{\text{opt}} = 1.93 \times 10^{-2} C^{0.485}, \quad D_{\text{opt}} = 4.64 C^{0.531}. \quad (2)$$

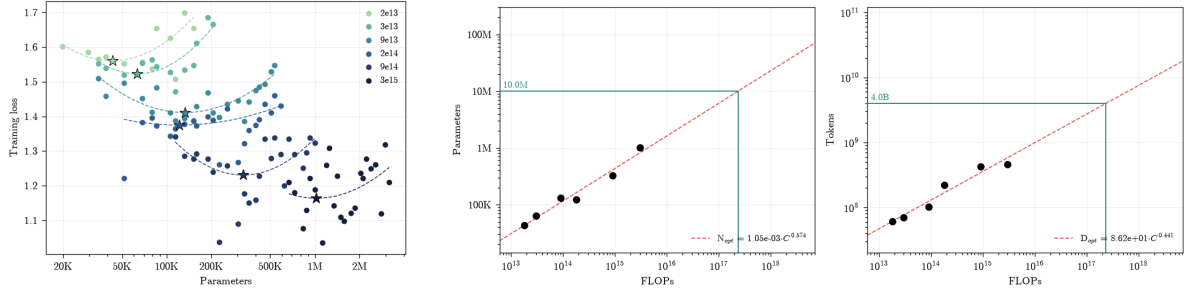


**Figure 1:** Approach 1. Left: training curves coloured by model size; grey points mark the lower envelope. Centre and right: model size and token count of the envelope, with power-law fits. The teal projection marks where the fit predicts a 10M-parameter model becomes compute-optimal, at  $C \approx 9.3 \times 10^{17}$  FLOPs and 16.2B tokens.

### 3 Approach 2: IsoFLOP profiles (dense)

In the second approach I fix eight training FLOP budgets from  $1.8 \times 10^{13}$  to  $3 \times 10^{15}$  and, for each budget, train a curated range of model sizes exactly to that budget. The number of steps for each model follows from the budget and its per-token cost, and the cosine schedule is set to match that horizon. For each budget I fit the final smoothed loss against  $\log N$  with a parabola and take the vertex as  $N_{\text{opt}}$ ;  $D_{\text{opt}}$  follows from the budget. A power law through the vertices gives the frontier (Figure 2).

I deviate from the paper in two ways. First, I reject grokked runs before fitting. For each budget I fit the parabola to all points by least squares and compute the standard deviation  $\sigma$  of its residuals. Runs whose loss sits more than  $2\sigma$  below the fitted curve are dropped and the parabola is refit, for at most three iterations and never below four remaining points. The rejection is one-sided because the two kinds of outlier have different causes. A run below the bulk is one that already grokked, and it would drag the vertex toward itself; a run above the curve is an ordinary under-trained model and is legitimate data. In practice the rule removes between zero and four runs per budget out of 12 to 24, so its effect is limited to removing the grokked tail. Second, I exclude two budgets before fitting. At  $C = 3 \times 10^{14}$  the fitted vertex lands at the far-left edge of the sampled model range, which means the minimum was not actually located. At  $C = 1.8 \times 10^{15}$  the fitted parabola is concave and has no minimum at all. Six budgets remain (Table 2).



**Figure 2:** Approach 2. Left: isoFLOP profiles;  $\star$  marks each fitted optimum. Centre and right: the frontier through the vertices. The 10M-parameter projection from this fit is  $C \approx 2.4 \times 10^{17}$  FLOPs and 4.0B tokens.

The fitted frontier is

$$N_{\text{opt}} = 1.05 \times 10^{-3} C^{0.574}, \quad D_{\text{opt}} = 86.2 C^{0.441}. \quad (3)$$

### 4 Approach 3: fitting a parametric loss function (dense)

In the third approach I fit every run’s final smoothed loss with the risk decomposition proposed in the paper,

$$\hat{L}(N, D) = E + \frac{A}{N^\alpha} + \frac{B}{D^\beta}, \quad (4)$$

| $C$ (train FLOPs)    | runs | $N_{\text{opt}}$ | $D_{\text{opt}}$  | curvature |
|----------------------|------|------------------|-------------------|-----------|
| $1.8 \times 10^{13}$ | 12   | 43.0K            | $6.1 \times 10^7$ | 0.079     |
| $3 \times 10^{13}$   | 13   | 63.3K            | $7.0 \times 10^7$ | 0.098     |
| $9 \times 10^{13}$   | 23   | 132.8K           | $1.0 \times 10^8$ | 0.057     |
| $1.8 \times 10^{14}$ | 22   | 121.8K           | $2.2 \times 10^8$ | 0.023     |
| $9 \times 10^{14}$   | 24   | 328.4K           | $4.2 \times 10^8$ | 0.086     |
| $3 \times 10^{15}$   | 23   | 1.0M             | $4.6 \times 10^8$ | 0.069     |

**Table 2:** Per-budget isoFLOP fits for the dense family, after the window filter and exclusions.

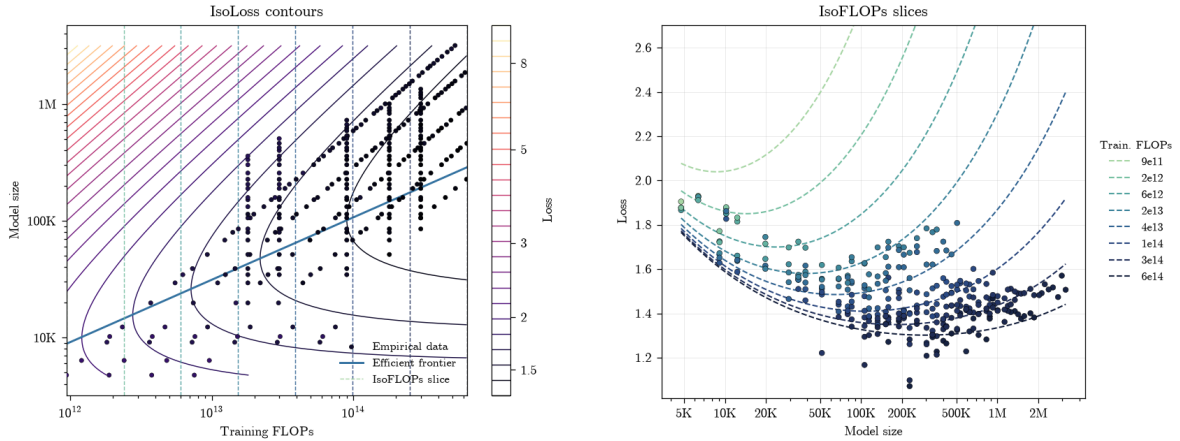
minimising a Huber loss ( $\delta = 10^{-3}$ ) on the log-residuals with L-BFGS and selecting the best fit from a grid of initialisations. This is the paper’s estimator, unchanged. The closed form of the constrained optimum gives  $a = \beta/(\alpha + \beta)$  and  $b = \alpha/(\alpha + \beta)$ .

I fit only the runs with  $C \leq 6.6 \times 10^{14}$ , which keeps 306 of the 463. The smooth surface model is false above the grokking transition, which Appendix A locates at roughly  $7 \times 10^{14}$ .

The fitted parameters are

$$E = 1.116, \quad A = 30.1, \quad B = 2433, \quad \alpha = 0.455, \quad \beta = 0.522 \Rightarrow a = 0.534, \quad b = 0.466. \quad (5)$$

The 10M-parameter projection from this fit is  $C \approx 4.9 \times 10^{17}$  FLOPs and 7.5B tokens.



**Figure 3:** The parametric fit. Left: isoloss contours with the closed-form efficient frontier. Right: isoFLOP slices over the raw runs.

The fitted value of  $E$  requires some interpretation. In Chinchilla,  $E$  is the entropy of natural text and is a real floor. This task is deterministic, so its true entropy is zero. The fitted  $E \approx 1.12$  is instead the level that pre-grokking learning converges to; it is the loss the surface predicts in the limit of unlimited model size and data. Random guessing over the ten digits costs  $\ln 10 \approx 2.30$  nats per answer digit, so this level corresponds to capturing roughly half of the answer entropy. That half is the part of the product that can be predicted without the full algorithm, for example the non-uniform digit statistics of products and the trailing digits, which depend only on the trailing digits of the operands. Grokked runs fall below  $E$ , which is why the surface cannot describe them and why the compute cut is needed. The fitted law is therefore a scaling law for this partial-structure phase; it describes how efficiently models learn the predictable part of the task, and it does not predict when the jump to the full algorithm occurs.

## 5 Dense results

All three estimates lie between 0.48 and 0.58, close to equal scaling and far from the allocation of Kaplan et al. (2020). Their fit scales the model much faster than the data ( $a = 0.73$  against  $b = 0.27$ ); under it, a  $10\times$  increase in compute grows the model  $5.5\times$  but the data only  $1.8\times$ . The spread between approaches is larger than in the paper. This is expected, since the fits here rest on 582 small and noisy runs, while theirs used over 400 much smoother ones.

| approach                        | $a$ ( $N_{\text{opt}} \propto C^a$ ) | $b$ ( $D_{\text{opt}} \propto C^b$ ) |
|---------------------------------|--------------------------------------|--------------------------------------|
| 1. Minimum over training curves | 0.485                                | 0.531                                |
| 2. IsoFLOP profiles             | 0.574                                | 0.441                                |
| 3. Parametric fit of the loss   | 0.534                                | 0.466                                |
| Chinchilla (paper)              | 0.50 / 0.49 / 0.46                   | 0.50 / 0.51 / 0.54                   |
| Kaplan et al. (2020)            | 0.73                                 | 0.27                                 |

**Table 3:** Dense scaling exponents against the reference values.

| target $N$ | A1: $C / D$                                           | A2: $C / D$                                          | A3: $C / D$                                          |
|------------|-------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|
| 100K       | $7.0 \times 10^{13} / 104\text{M}$                    | $7.8 \times 10^{13} / 116\text{M}$                   | —                                                    |
| 1M         | $8.1 \times 10^{15} / 1.3\text{B}$                    | $4.3 \times 10^{15} / 678\text{M}$                   | —                                                    |
| <b>10M</b> | <b><math>9.3 \times 10^{17} / 16.2\text{B}</math></b> | <b><math>2.4 \times 10^{17} / 4.0\text{B}</math></b> | <b><math>4.9 \times 10^{17} / 7.5\text{B}</math></b> |

**Table 4:** Projected compute-optimal allocations. The  $4\times$  spread at 10M reflects the uncertainty of extrapolating two decades beyond the data; Chinchilla faced a similar spread when choosing between 40B and 70B parameters for their validation model.

## 6 Mixture of experts

I repeat the three approaches on the MoE family with the same estimators and the same cuts unless noted. The family has 119 runs over 13 sizes with top-1-of-4 experts. It covers a 1-in-4 subset of the dense sizes rather than all 55, because the expert dispatch makes MoE steps slower at these scales and a full sweep would have taken considerably more wall-clock time. Every fit is reported against two parameter counts. *Active* parameters count only the routed expert and match the dense family’s  $N$  and FLOPs per token. *Total* parameters count all four experts.

### 6.1 Approach 1

The envelope method needs many model sizes competing at every budget, and the censoring rule of Section 2 drops any budget reached by fewer than 10 of them. With 13 sizes and four horizons, budgets above  $C = 2.2 \times 10^{14}$  are reached by fewer than 10 sizes, so the fixed-horizon runs alone support an envelope of less than two decades of compute, which is too short to fit a slope reliably. I therefore include the runs from approach 2 in the approach-1 analysis, so the 67 fixed-budget runs contribute their training curves to the same minimum alongside the 52 fixed-horizon runs. This extends the envelope to  $3.0 \times 10^{15}$ . The cost is that an included run often wins a budget in the middle of its training rather than near the end, so only 20.8% of envelope points come from the last 15% of a run.

$$\text{active: } N_{\text{opt}} = 1.48 \times 10^{-4} C^{0.639}, \quad D_{\text{opt}} = 552 C^{0.380}. \quad (6)$$

The steep 0.64 should be interpreted with caution. With 13 sizes the envelope is a staircase with roughly 5 steps, and the slope is dominated by the largest jump. Approaches 2 and 3 give more reliable estimates for the MoE family.

### 6.2 Approach 2

Two budgets are excluded for the same edge-minimum failure as the dense  $3 \times 10^{14}$ , namely  $C = 9 \times 10^{14}$  and  $1.8 \times 10^{15}$ . Six budgets remain.

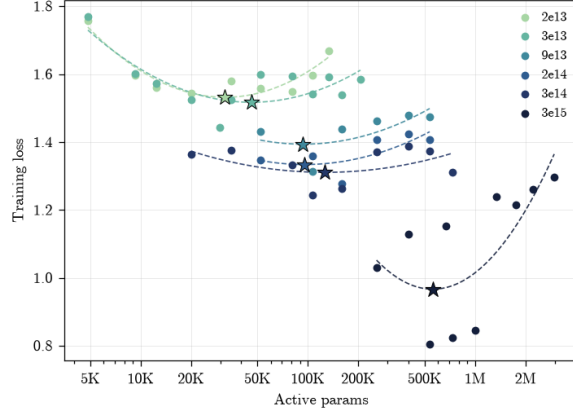
$$\text{active: } N_{\text{opt}} = 2.56 \times 10^{-3} C^{0.536}, \quad D_{\text{opt}} = 35.0 C^{0.480}; \quad \text{total: } N_{\text{opt}} \propto C^{0.540}. \quad (7)$$

### 6.3 Approach 3

The fit uses the same  $C \leq 6.6 \times 10^{14}$  cut and keeps 80 runs.

$$\text{active: } E = 1.139, A = 51.9, B = 2298, \alpha = 0.538, \beta = 0.515 \Rightarrow a = 0.489; \quad (8)$$

$$\text{total: } E = 1.137, A = 81.6, B = 2231, \alpha = 0.523, \beta = 0.513 \Rightarrow a = 0.495. \quad (9)$$



**Figure 4:** MoE isoFLOP profiles, plotted against active parameters.

| approach                        | $a$ (active) | $b$ (active) | $a$ (total) |
|---------------------------------|--------------|--------------|-------------|
| 1. Minimum over training curves | 0.639        | 0.381        | 0.641       |
| 2. IsoFLOP profiles             | 0.536        | 0.480        | 0.540       |
| 3. Parametric fit of the loss   | 0.489        | 0.511        | 0.495       |

**Table 5:** MoE scaling exponents.

## 7 Discussion

All three approaches, applied to both model families, give similar estimates of the optimal allocation. Parameter count and training tokens should be scaled in roughly equal proportions, with fitted exponents between 0.49 and 0.57 for the dense family. This matches the conclusion of Hoffmann et al. (2022) despite the much smaller scale and the change of task.

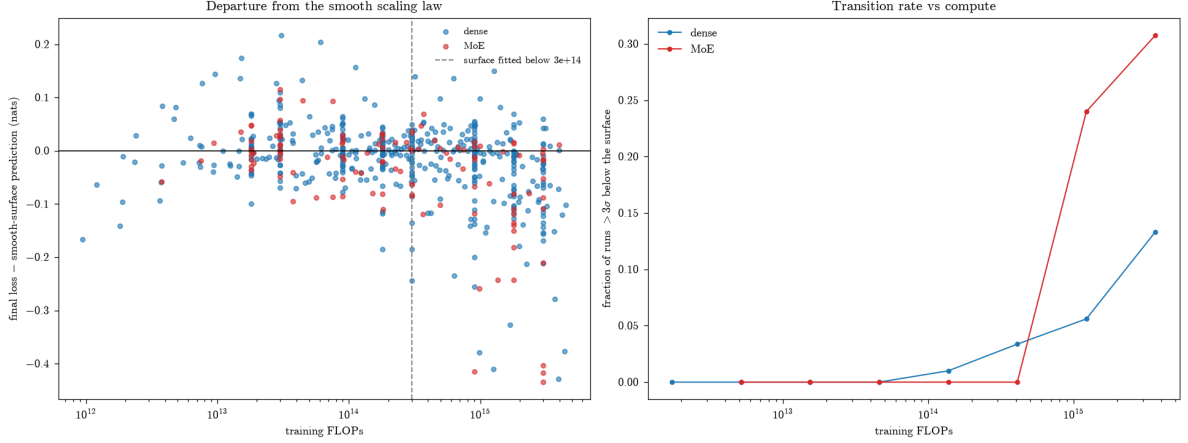
The MoE results suggest that sparsity does not change this allocation. By approaches 2 and 3, the MoE frontier has the same exponent as the dense frontier, whether measured in active or in total parameters, and the difference between the families appears in the constant instead. At a given budget the optimal MoE model tracks the dense optimum in active parameters, holds about 3.2 times more total parameters, and reaches a slightly lower loss (0.97 against roughly 1.16 at  $C = 3 \times 10^{15}$ ).

This comparison should be interpreted with some care. The budgets where the MoE advantage is largest are also the budgets where grokking is most frequent. Above roughly  $7 \times 10^{14}$  FLOPs, 24 to 31% of MoE runs per compute bin fall more than  $3\sigma$  below the smooth surface, compared to 6 to 13% of dense runs (Appendix A). Some of the MoE advantage at these budgets therefore comes from runs that have left the regime the fitted law describes. The higher transition rate may itself be an effect of the additional total capacity, but I did not investigate this further.

Finally, the fragile part of this analysis was not the allocation estimate but the assumptions behind it. The loss needs to fall smoothly with compute and to remain a meaningful measure of ability over the measured range, and on arithmetic tasks neither can be taken for granted. The fitted  $E$  is the floor of the pre-grokking phase rather than the entropy of the data, and the fitted surface does not indicate where the transition will occur. In this setting, most of the work of deriving the scaling law went into choosing a task on which its assumptions hold.

## A The grokking transition

I fit the approach-3 surface on the runs below  $C = 3 \times 10^{14}$  only and measure every run’s residual against it. A run more than  $3\sigma$  below the surface has left the smooth regime, and I call it transitioned. I report the rate of transitioned runs per compute bin rather than the first crossing, because the dense family has about  $4\times$  more runs and would always produce the earliest grokked run by sampling alone.



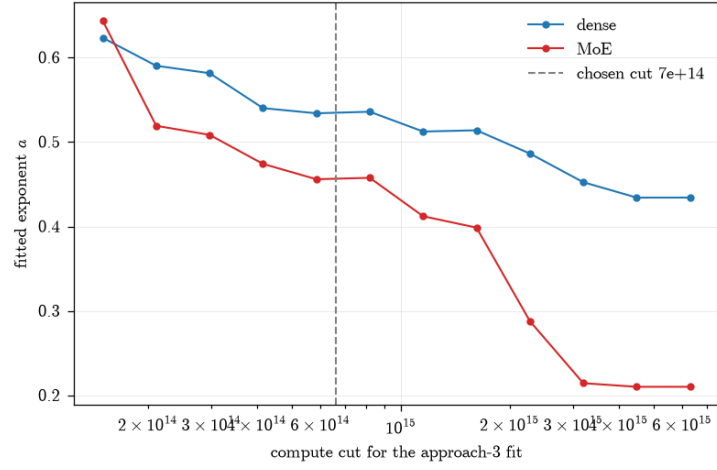
**Figure 5:** Left: residuals against the smooth surface fitted below  $3 \times 10^{14}$ . Right: fraction of runs more than  $3\sigma$  below the surface, per compute bin.

| compute bin                                 | dense runs | dense rate | MoE runs | MoE rate |
|---------------------------------------------|------------|------------|----------|----------|
| $7.9 \times 10^{13}$ – $2.4 \times 10^{14}$ | 100        | 1%         | 23       | 0%       |
| $2.4 \times 10^{14}$ – $7.1 \times 10^{14}$ | 89         | 3%         | 22       | 0%       |
| $7.1 \times 10^{14}$ – $2.1 \times 10^{15}$ | 107        | 6%         | 25       | 24%      |
| $2.1 \times 10^{15}$ – $6.3 \times 10^{15}$ | 45         | 13%        | 13       | 31%      |

**Table 6:** Transition rates. Bins below  $7.9 \times 10^{13}$  are 0% everywhere. The fitted surfaces are  $E=1.119$ ,  $\alpha=0.442$ ,  $\beta=0.536$ ,  $\sigma=0.055$  on 264 dense runs and  $E=1.127$ ,  $\alpha=0.530$ ,  $\beta=0.478$ ,  $\sigma=0.047$  on 71 MoE runs.

The onset sits at roughly  $7 \times 10^{14}$  for both families, and the approach-3 cut of  $6.6 \times 10^{14}$  is chosen just below it. The largest residuals reach  $-0.44$  nats, far larger than the residual noise of the fits ( $\sigma \approx 0.05$ ). Small models sometimes grok ahead of larger ones, which also roughens the approach-1 envelope at high compute, where an early-grokking small model can briefly hold the minimum against every larger size. These measurements are for 6-digit multiplication; with 3-digit addition the transition happens so early that no smooth window exists at all, which is why the task was changed.

Figure 6 shows how this choice of cut affects the approach-3 result. I refit the surface using only the runs below a threshold and plot the fitted exponent  $\alpha$  against that threshold. Below roughly  $3 \times 10^{14}$  the estimates are high and unstable, since few runs remain and they cover a narrow compute range. Around the chosen cut the dense exponent is stable near 0.53, and the MoE exponent lies between 0.46 and 0.51, moving by a few hundredths between nearby thresholds because of its smaller run count. As the threshold moves past the transition and grokked runs enter the fit, both exponents fall, to 0.43 for dense and 0.21 for MoE on the full data. The MoE fit degrades more because a larger fraction of its high-compute runs have transitioned.

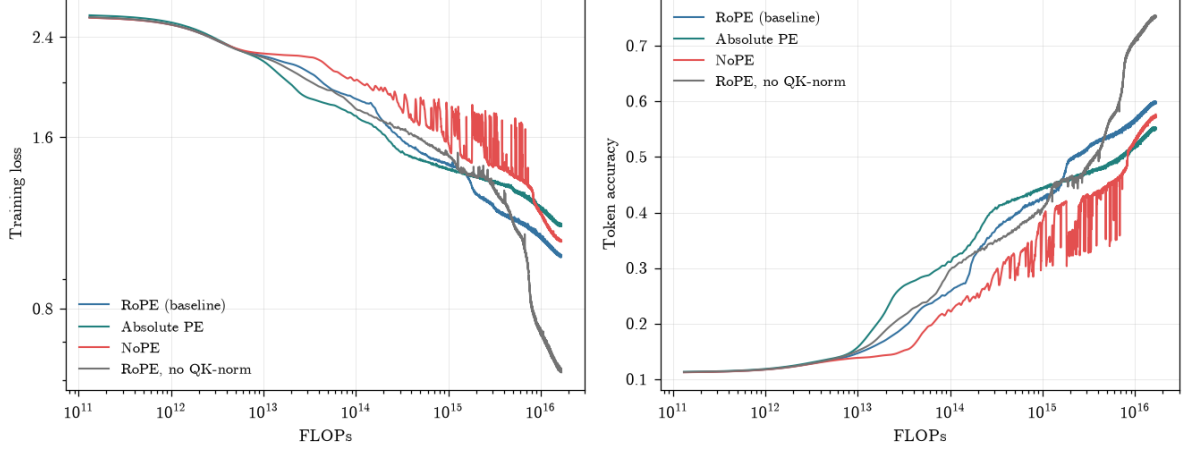


**Figure 6:** The fitted exponent  $a$  of the approach-3 surface as a function of the compute cut. Each point refits the surface on the runs below that threshold. The dashed line marks the cut used in this report.



## B Architecture ablations

The model family fixes two attention details that the main sweeps never vary: RoPE for position information and QK-norm. To check how much these choices matter, I trained the largest dense model (3.17M parameters) under four variants, one per chip of a TPU v5e-4. The baseline keeps RoPE and QK-norm. The other three variants replace RoPE with a learned absolute position embedding, remove position information entirely (NoPE), and keep RoPE but remove QK-norm. Every other setting matches the main runs, including the seed. Each variant trains for 128,000 steps, four times the longest horizon the 3.17M model sees in the main sweeps, reaching  $C = 1.7 \times 10^{16}$ .



**Figure 7:** Training loss (left) and token accuracy over the 12 answer digits (right) for four variants of the 3.17M model, each trained for 128,000 steps. Curves are smoothed with the same Gaussian filter as the main analysis ( $\sigma = 20$  steps).

Figure 7 shows the result. In the smooth regime the variants behave similarly. At  $C = 1 \times 10^{15}$  the four losses lie between 1.40 and 1.56. NoPE is consistently the highest and is the only variant that shows instability, in the form of repeated loss spikes concentrated between roughly  $1 \times 10^{15}$  and  $7 \times 10^{15}$  FLOPs. The variants separate at the transition. The run without QK-norm transitions first: its loss crosses 1.0 at  $C = 6.9 \times 10^{15}$  and falls to 0.62 by the end of training, with token accuracy reaching 75%. The same run is not fully stable. It shows small loss spikes between roughly  $1 \times 10^{15}$  and  $7 \times 10^{15}$  FLOPs, much lower in height than the NoPE spikes and absent from the other two variants. The baseline crosses 1.0 at  $1.5 \times 10^{16}$ , close to the end of its horizon. This is consistent with the main sweeps, where the 3.17M model had not transitioned by  $4.2 \times 10^{15}$ . Neither position-embedding replacement transitions within the horizon. Absolute position embeddings end at a loss of 1.12, matching the fitted floor  $E$ , and NoPE ends at 1.05. Exact-match accuracy over the full product stays below 1% for every variant, so even the transitioned runs remain far from solving the task at this compute.

Two observations seem worth stating. First, these architecture choices matter here mainly through the timing of the transition rather than through the smooth regime, where the four curves differ by at most about 0.2 nats. Second, removing QK-norm brings the transition forward by about a factor of 2 in compute. The transition threshold of Appendix A is therefore specific to this architecture, and a different baseline could move it.

## C FLOP computation

As in Appendix F of the paper, I include all operations in the forward pass and count a multiply-accumulate as 2 FLOPs. Unlike the generic rule, I also count normalisations, activations, and residual additions, which are not negligible at these sizes. With  $S = 26$  (`seq_len`),  $V = 12$  (`vocab_size`),  $D$  (`d_model`),  $F$  (`d_ffw`),  $L$  (`n_layers`),  $N_q$  query heads,  $N_{kv}$  key/value heads, and  $H = 8$  (`d_head`):

- Embeddings:  $2SVD$
- Attention (single layer):
  - Q, K, V projections:  $2SDH(N_q + 2N_{kv})$
  - QK-RMSNorm and RoPE:  $4SN_qH + 4SN_{kv}H + 3S(N_q + N_{kv})H + 3SH$
  - key @ query logits:  $2S^2HN_q$
  - scaling and softmax:  $S^2N_q + 5N_qS^2$
  - softmax @ value reductions:  $2S^2HN_q$
  - output projection:  $2S(HN_q)D$
- SwiGLU dense block (single layer, 3 matmuls):  $6SDF + 6SF$
- Pre-attention and pre-MLP RMSNorm:  $2 \times 4SD$ ; residual additions:  $2SD$
- Final RMSNorm:  $4SD$ ; final logits:  $2SVD$

$$F_{\text{fwd}} = 2SVD + L(\text{attn} + \text{mlp} + 10SD) + 4SD + 2SVD, \quad \text{FLOPs/token} = F_{\text{fwd}}/S, \quad (10)$$

with backward  $\approx 2 \times$  forward, so training FLOPs per token are  $3 \times F_{\text{fwd}}/S$ . As in the paper, the difference against the  $C = 6ND$  rule is small and shrinks with scale; their Table A4 approaches 1.0 for billion-parameter models. Table 7 shows the small end of the same trend. The ratio is driven by the  $O(S^2)$  attention terms, the embeddings, and the normalisation being relatively larger in small models.

| params | FLOPs/token / $6N$ |
|--------|--------------------|
| 19.8K  | 1.17               |
| 105.9K | 1.11               |
| 533.7K | 1.07               |
| 3.17M  | 1.03               |

**Table 7:** The exact per-token count against the  $6N$  rule.

## D Fitting details and exclusions

- **Smoothing.** A Gaussian filter with  $\sigma = 20$  steps is applied to every loss curve before any use.
- **Envelope censoring (approach 1).** Grid points where fewer than 10 distinct sizes reach the budget are dropped. Dense keeps  $C \leq 3.4 \times 10^{15}$  and MoE keeps  $C \leq 3.0 \times 10^{15}$ .
- **Trim fit (approach 2).** The parabola is refit iteratively while dropping points more than  $2\sigma$  below the curve, for at most 3 iterations and never below 4 points. The rejection is one-sided because grokking only ever lowers loss.
- **Per-budget model windows (approach 2, dense).** Each budget fits only the sizes inside a hand-curated window (Appendix E) so that every parabola brackets its minimum. 117 of 153 remaining runs pass the windows.
- **Excluded budgets.** Dense  $3 \times 10^{14}$  (vertex at the range edge), dense  $1.8 \times 10^{15}$  (concave), MoE  $9 \times 10^{14}$  and  $1.8 \times 10^{15}$  (vertex at the range edge).
- **Approach-3 cut.**  $C \leq 6.6 \times 10^{14}$ , from Appendix A.
- Bootstrap confidence intervals on  $(a, b)$ , as in Table 2 of the paper, are not computed. They would strengthen the exponent tables.

## E Model ranges per FLOP budget

| $C$ (train FLOPs)    | dense window ( $N$ )      | MoE range (active $N$ )   |
|----------------------|---------------------------|---------------------------|
| $1.8 \times 10^{13}$ | 19.8K–151.6K              | 4.8K–132.9K               |
| $3 \times 10^{13}$   | 34.6K–205.5K              | 4.8K–206.5K               |
| $9 \times 10^{13}$   | 34.6K–533.7K              | 51.8K–535.8K              |
| $1.8 \times 10^{14}$ | 51.5K–590.1K              | 51.8K–535.8K              |
| $3 \times 10^{14}$   | 85.2K–727.3K <sup>†</sup> | 20.0K–729.6K              |
| $9 \times 10^{14}$   | 114.0K–1.00M              | 159.3K–1.01M <sup>†</sup> |
| $1.8 \times 10^{15}$ | 506.1K–2.03M <sup>†</sup> | 106.5K–2.0M <sup>†</sup>  |
| $3 \times 10^{15}$   | 666.7K–3.17M              | 257.7K–3.0M               |

**Table 8:** Model sizes entering each isoFLOP fit. <sup>†</sup> marks budgets excluded from the frontier fit, for the reasons in Appendix D.

## F Model zoo

All models use  $d_{\text{ffw}} = 4d_{\text{model}}$ ,  $d_{\text{head}} = 8$ , and  $n_{\text{heads}} = n_{kv} = d_{\text{model}}/8$ . FLOPs/token is the exact forward count of Appendix C. MoE models share the dense backbone with 4 experts per MLP and top-1 routing. The initial learning rate of each size was selected from three options, 1e-2, 3e-3, and 1e-3, by running a short training run with each and keeping the one with the lowest loss.

| params | $L$ | $d_{\text{model}}$ | LR   | FLOPs/token | params | $L$ | $d_{\text{model}}$ | LR   | FLOPs/token |
|--------|-----|--------------------|------|-------------|--------|-----|--------------------|------|-------------|
| 4.76K  | 1   | 16                 | 1e-2 | 11,792      | 506.1K | 6   | 72                 | 3e-3 | 1,073,304   |
| 6.36K  | 1   | 16                 | 1e-2 | 15,056      | 533.7K | 8   | 64                 | 3e-3 | 1,139,904   |
| 9.12K  | 2   | 16                 | 1e-2 | 22,752      | 590.1K | 7   | 72                 | 3e-3 | 1,251,564   |
| 10.2K  | 1   | 24                 | 1e-2 | 23,820      | 623.7K | 6   | 80                 | 3e-3 | 1,315,424   |
| 12.3K  | 2   | 16                 | 1e-2 | 29,280      | 666.7K | 10  | 64                 | 3e-3 | 1,424,048   |
| 19.8K  | 2   | 24                 | 1e-2 | 46,392      | 727.3K | 7   | 80                 | 3e-3 | 1,533,968   |
| 29.4K  | 3   | 24                 | 1e-2 | 68,964      | 830.9K | 8   | 80                 | 3e-3 | 1,752,512   |
| 34.6K  | 2   | 32                 | 1e-2 | 78,224      | 878.9K | 7   | 88                 | 3e-3 | 1,845,044   |
| 39.0K  | 4   | 24                 | 1e-2 | 91,536      | 926.3K | 11  | 72                 | 3e-3 | 1,964,604   |
| 51.5K  | 3   | 32                 | 1e-2 | 116,504     | 1.00M  | 8   | 88                 | 3e-3 | 2,107,968   |
| 68.3K  | 4   | 32                 | 1e-2 | 154,784     | 1.13M  | 9   | 88                 | 3e-3 | 2,370,892   |
| 79.7K  | 3   | 40                 | 1e-2 | 176,332     | 1.19M  | 8   | 96                 | 1e-3 | 2,496,192   |
| 85.2K  | 5   | 32                 | 1e-2 | 193,064     | 1.25M  | 10  | 88                 | 1e-3 | 2,633,816   |
| 105.9K | 4   | 40                 | 1e-2 | 234,416     | 1.34M  | 9   | 96                 | 1e-3 | 2,807,592   |
| 114.0K | 3   | 48                 | 1e-2 | 248,448     | 1.42M  | 7   | 112                | 1e-3 | 2,950,304   |
| 132.1K | 5   | 40                 | 1e-2 | 292,500     | 1.49M  | 10  | 96                 | 1e-3 | 3,118,992   |
| 151.6K | 4   | 48                 | 1e-2 | 330,432     | 1.57M  | 9   | 104                | 1e-3 | 3,281,156   |
| 158.3K | 6   | 40                 | 1e-2 | 350,584     | 1.75M  | 10  | 104                | 1e-3 | 3,645,128   |
| 189.2K | 5   | 48                 | 1e-2 | 412,416     | 1.86M  | 8   | 120                | 1e-3 | 3,857,472   |
| 205.5K | 4   | 56                 | 1e-2 | 442,832     | 2.03M  | 10  | 112                | 1e-3 | 4,212,224   |
| 226.8K | 6   | 48                 | 1e-2 | 494,400     | 2.12M  | 8   | 128                | 1e-3 | 4,376,768   |
| 256.6K | 5   | 56                 | 1e-2 | 552,812     | 2.23M  | 11  | 112                | 1e-3 | 4,632,864   |
| 302.0K | 8   | 48                 | 3e-3 | 658,368     | 2.38M  | 9   | 128                | 1e-3 | 4,923,032   |
| 334.2K | 5   | 64                 | 3e-3 | 713,688     | 2.56M  | 11  | 120                | 1e-3 | 5,301,684   |
| 358.6K | 7   | 56                 | 3e-3 | 772,772     | 2.79M  | 12  | 120                | 1e-3 | 5,783,088   |
| 400.7K | 6   | 64                 | 3e-3 | 855,760     | 2.97M  | 17  | 104                | 1e-3 | 6,192,932   |
| 422.0K | 5   | 72                 | 3e-3 | 895,044     | 3.17M  | 12  | 128                | 1e-3 | 6,561,824   |
| 460.7K | 9   | 56                 | 3e-3 | 992,732     |        |     |                    |      |             |

**Table 9:** The dense family (55 models).

| active params | total params | $L$ | $d_{\text{model}}$ | LR   | FLOPs/token |
|---------------|--------------|-----|--------------------|------|-------------|
| 20.0K         | 62.8K        | 2   | 24                 | 1e-2 | 46,882      |
| 51.8K         | 165.0K       | 3   | 32                 | 1e-2 | 117,479     |
| 106.5K        | 341.3K       | 4   | 40                 | 1e-2 | 236,036     |
| 159.3K        | 511.4K       | 6   | 40                 | 1e-2 | 353,014     |
| 257.7K        | 829.8K       | 5   | 56                 | 1e-2 | 555,637     |
| 402.2K        | 1.30M        | 6   | 64                 | 3e-3 | 859,630     |
| 535.8K        | 1.73M        | 8   | 64                 | 3e-3 | 1,145,064   |
| 729.6K        | 2.36M        | 7   | 80                 | 3e-3 | 1,539,603   |
| 1.01M         | 3.26M        | 8   | 88                 | 3e-3 | 2,115,048   |
| 1.35M         | 4.36M        | 9   | 96                 | 1e-3 | 2,816,277   |
| 1.75M         | 5.67M        | 10  | 104                | 1e-3 | 3,655,578   |
| 2.23M         | 7.23M        | 11  | 112                | 1e-3 | 4,645,239   |
| 2.98M         | 9.65M        | 17  | 104                | 1e-3 | 6,210,697   |

**Table 10:** The MoE family (13 models, a 1-in-4 subset of the dense sizes). FLOPs/token counts the routed expert and the router.